

CRITICAL EXPOSITION: HOW TWO-LAYER NET- WORKS LEARN IN ONE (GIANT) GRADIENT STEP

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Computational Methods for Inverse Problems

ABSTRACT

This report studies **How Two-Layer Neural Networks Learn, One (Giant) Step at a Time** (Dandi et al., 2024) (JMLR, 2024). The paper analyzes early training of a two-layer network on Gaussian inputs with multi-index labels, where all low-dimensional structure is encoded in a link function over unknown teacher projections. The core question is when a small number of large-batch gradient steps on first-layer weights yields nontrivial alignment with the teacher subspace V^* , thus departing from the initialization (lazy) kernel.

Following (Dandi et al., 2024), I separate three ingredients—batch size n , number of gradient steps, and target structure (Hermite / leap / staircase)—and stress that they are not interchangeable. In one giant full-batch step, batch scaling $n = \Theta(d)$ supports alignment but only to a single teacher direction unless n reaches $\Theta(d^2)$ scale for multi-direction specialization; still harder directions may demand $n = \Theta(d^\ell)$ with leap index ℓ . With multiple steps and fresh batches of size $n = \Theta(d)$, targets satisfying a staircase coupling can progressively recruit several directions over time, while obstructed pieces retain $\Theta(d^2)$ bottlenecks. Finally, (Dandi et al., 2024) link these geometric facts to approximation and generalization relative to the initialization kernel—including settings where one gradient step helps or fails versus frozen features.

Critical discussion of assumptions, limitations, and preliminary EE592 experiments is in Section 2; the technical overview begins in Section 1.

AI disclosure: Generative AI tools were used for drafting and ideation; the author edited the text and takes responsibility for all content.

1 SETTING AND CONTRIBUTIONS

Unless noted, asymptotic statements match (Dandi et al., 2024) (their §1–2; large- d scaling).

1.1 NOTATION AT A GLANCE

Symbol	Meaning
d	Input dimension ($z \in \mathbb{R}^d$).
r	Number of teacher directions.
n	Batch size / samples per update stage.
p	Network width (hidden units).
$W \in \mathbb{R}^{p \times d}$	First-layer matrix; row w_i is neuron- i direction.
$a \in \mathbb{R}^p$	Second-layer vector.
W^*	Teacher matrix (rows w_j^*).
f^*, g^*	Target: $f^*(z) = g^*(W^*z)$.
V^*	Teacher span $\text{span}\{w_1^*, \dots, w_r^*\}$.
η	Learning rate for W updates.
ℓ	Leap index (first Hermite order with signal).

$\Theta(\cdot)$	Asymptotic scaling notation.
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1.2 PROBLEM SETUP

Data and target. Covariates $z \propto \mathcal{N}(0, I_d)$ in $\mathbb{R}^d$. Labels follow a multi-index representation: orthogonal teachers $w_1^*, \dots, w_r^*$ and a link $g^* : \mathbb{R}^r \rightarrow \mathbb{R}$ with

$$y = f^*(z) := g^*(\langle w_1^*, z \rangle, \dots, \langle w_r^*, z \rangle). \quad (1)$$

Isotropic inputs put **all** exploitable structure in f^* . That modeling choice is deliberate: classical kernels depend on isotropic geometry of z but do not automatically exploit hidden low-dimensional structure in y , so improvements over kernel behavior here are evidence of **representation learning** rather than an artifact of input clustering (Dandi et al., 2024).

Two-layer predictor. First- and second-layer parameters $W \in \mathbb{R}^{p \times d}$, $a \in \mathbb{R}^p$, and activation σ ,

$$\hat{f}(z; W, a) = \frac{1}{\sqrt{p}} \sum_{i=1}^p a_i \sigma(\langle w_i, z \rangle). \quad (2)$$

Training regime analyzed. (Dandi et al., 2024) focus on the **early** phase: one or more full-batch gradient steps on W under squared loss, with batch size n coupled to d (typically $n = \Theta(d)$ versus $\Theta(d^2)$ comparisons) and learning-rate scaling tied to width p as in their theorems. For multi-step analyses they use **fresh i.i.d. batches** each iteration and (like related work) treat second-layer training separately—idealizations that simplify the Hermite and concentration analysis (Dandi et al., 2024).

Operational notion of learning. Beating the kernel at initialization means growing overlap between rows of W and $V^* := \text{span}\{w_1^*, \dots, w_r^*\}$. Whether overlap arises, and in how many orthogonal directions, is driven jointly by (n, steps) and the **Hermite staircase** structure of g^* (Dandi et al., 2024).

1.3 MAIN CONTRIBUTIONS

1.3.1 A SINGLE ORGANIZING TENSION

(Dandi et al., 2024) argue that **specialization**—encoding distinct teacher directions in distinct hidden units—is not a monotone consequence of more data.’ Their comparison is between one full-batch update built from n samples (each sample contributes an additive term to the gradient; those terms are independent and can be farmed out to parallel workers, then averaged) and **classical one-pass SGD**, where updates happen one after another so later steps depend on earlier ones. Thus needing a large n for one full batch is not the same as needing many **serial** gradient steps: **sample counts** and **wall-clock complexity** (elapsed real time for the computation to finish, as opposed to an abstract count of serial operations or steps) need not track one another; what matters for their theory is which directions appear in the Hermite expansion of g^* at an order detectable by one batch step at that (n, step) pair (Dandi et al., 2024) (their leap index and staircase definitions formalize this).

1.3.2 ONE FULL-BATCH GRADIENT STEP ON W

Their Theorems 4–5 and Fig. 1 summarize a batch-size hierarchy for one step: alignment already requires n at least on $\Theta(d)$ order with η coupled to p , yet **at most one** direction in V^* can emerge at that scale when $r > 1$. Acquiring several directions **simultaneously** in one update pushes the batch to $\Theta(d^2)$. Even then, some directions can force $n = \Theta(d^\ell)$ where ℓ is the leap index (Definition 3): the first Hermite degree at which certain components of g^* appear (Dandi et al., 2024).

1.3.3 MULTIPLE STEPS: STAIRCASE STRUCTURE AND OBSTRUCTIONS

Theorem 7 and Fig. 2 describe a different regime: with repeated steps, **fresh** batches of size $n = \Theta(d)$ suffice to learn **multiple** directions over time **provided** new directions are coupled to previous ones through the staircase property (made precise in their §3.4). Intuitively, later directions become statistically predictable once earlier ones align—so the network can climb a sequence of linearized“ subproblems. Directions lacking this coupling (and with degenerate low Hermite mass) remain stuck behind the same $\Theta(d^2)$ barrier that already obstructs single-step multi-index learning (Dandi et al., 2024).

1.3.4 APPROXIMATION, GENERALIZATION, AND COMPARISON TO RANDOM FEATURES

Section 2.3 connects learned subspaces to what degree- k kernels could approximate at initialization (benchmarking against random-features scaling (Mei and Montanari, 2022), as discussed in (Dandi et al., 2024)). Fig. 3-style experiments show when one gradient step on W lowers test error relative to random features and when it does not: the deciding issue is whether the nonlinear signal has components along directions the first step can align with (Dandi et al., 2024).

1.4 COMPARISON TO THE REFERENCE PAPER

1.4.1 BA ET AL. (BA ET AL., 2022) AS NEIGHBOR WORK

(Ba et al., 2022) study one gradient step on the first layer of a two-layer network in a proportional high-dimensional limit: isotropic Gaussian inputs, single-index teacher, additive noise on y , and a sharp comparison between small versus large learning-rate regimes for representation improvement over the initial kernel (Ba et al., 2022). (Dandi et al., 2024) retain the same early feature learning“ spirit but change the target class, the noise model, and the algorithmic questions they prioritize (batch scaling and multiple steps); we spell out overlaps and splits below.

1.4.2 WHERE THE ASSUMPTIONS AGREE

- **Inputs.** Both works take features to be standard Gaussian (isotropic), so all nontrivial structure is encoded in the label map rather than in anisotropic covariates (Ba et al., 2022; Dandi et al., 2024).
- **Architecture and scaling.** Shallow two-layer networks with the usual $\frac{1}{\sqrt{p}}$ width normalization appear in both analyses (Ba et al., 2022; Dandi et al., 2024).
- **Training protocol (family resemblance).** Both separate learning of the first layer from fitting the second layer and use **fresh** batches for different phases (Ba et al. use two independent batches for the two layers in their one-step setup; (Dandi et al., 2024) use a fresh batch at each gradient step when analyzing multi-step training). In either case, decoupling layers avoids coupled finite-width pathologies and matches how this line of work models early full-batch GD on the first layer (Dandi et al., 2024).
- **Hermite viewpoint.** Each paper expands the target and loss geometry using Hermite analysis on Gaussian inputs; leap-index language enters when discussing which Hermite orders drive alignment (Ba et al., 2022; Dandi et al., 2024).

1.4.3 WHERE THEY DIFFER (AND WHAT LEAP INDEX“ MEANS IN EACH)

In (Ba et al., 2022) the teacher is **single-index**: $y = f^{*(x)} + \varepsilon$, with a scalar projection of x driving f^* . There the target has no constant Hermite component and already depends linearly on the teacher direction, so the leap index is $\ell = 1$ in (Dandi et al., 2024)‘s terminology: the first Hermite order at which that direction contributes is linear (Ba et al., 2022).

(Dandi et al., 2024) instead treat **multi-index** teachers $y = g^{*(W^*z)}$ with r orthonormal directions and **noiseless** labels. Their leap index ℓ is **not** fixed to 1: it indexes how deep

one must go in the Hermite expansion along a direction before the target’s dependence on that direction turns on.” When $\ell = 1$, a toy linear functional $w_1^T z$ has leap one (the minimal case); when $\ell > 1$, (Dandi et al., 2024) show that even enormous single-batch updates may still fail to learn that direction until n reaches $\Theta(d^\ell)$. Thus Ba et al.’s baseline single-index story aligns with the **simplest** leap-1 picture, whereas (Dandi et al., 2024) also treats cases with $\ell > 1$ and with several interacting directions through g^* (Dandi et al., 2024).

Summary. Similar isotropic-Gaussian + shallow-net setup; dissimilar **target complexity** (single- vs multi-index), **noise** (present vs absent), and **role of ℓ** (Ba et al. emphasize a leap-1 single-index tractable case; (Dandi et al., 2024) treat general ℓ and multi-step/staircase structure).

1.4.4 MAP BACK TO (DANDI ET AL., 2024) (SECTIONS / FIGURES)

Equations (1)–(2) in (Dandi et al., 2024) match our setup block above. Their abstract’s three bullets correspond to: §2.1 / Thms. 4–5 / Fig. 1 (single-step batch hierarchy and leap complexity); §2.2 / Thm. 7 / Fig. 2 (multi-step staircase learning at $n = \Theta(d)$); §2.3 / Fig. 3 (approximation and generalization versus initialization kernels, including comparison to random-features scaling (Mei and Montanari, 2022)). Section Section 1.3 tracks that order intentionally (Dandi et al., 2024).

Reading across papers. The progression from (Ba et al., 2022) to (Dandi et al., 2024) is not replace Ba’s theorem“ but change the task class and ask more detailed resource questions“: how large must a **single** batch be to unlock multiple directions in one update, how does leap ℓ obstruct that, and when do **several** large-batch steps at $n = \Theta(d)$ recover staircase targets that a single step cannot (Dandi et al., 2024).

2 CRITICAL DISCUSSION AND LIMITATIONS

This section summarizes scope conditions and limitations of (Dandi et al., 2024) and relates them to nearby deep-learning theory.

2.1 GAUSSIAN, NOISELESS, TEACHER-ONLY STRUCTURE

The analysis assumes isotropic Gaussian inputs and **noiseless** labels from a realized multi-index teacher (Dandi et al., 2024). This supports Hermite/Stein/Gaussian-equivalence arguments, but excludes correlated designs, heavy tails, covariate shift, and label noise. (Ba et al., 2022) use additive label noise in their single-index setting, so the two papers compare different statistical regimes. In practice, part of the observed error may be dominated by noise rather than by representation mismatch.

In related statistical and signal-processing settings, controlled noise can act as regularization (noise injection as Tikhonov smoothing) or improve detectability in nonlinear systems (stochastic resonance) (Bishop, 1995; Gammaitoni et al., 1998). Modern deep-learning analyses also study noise scale, batch size, and generalization in SGD-based training (Keskar et al., 2017; Mandt et al., 2017; Smith and Le, 2018).

Can moderate noise ever improve learning dynamics by regularizing alignment, rather than only increasing estimation error?

2.2 PROTOCOL: FRESH BATCHES, TWO STAGES, SYMMETRIZED INIT

(Dandi et al., 2024) study giant-batch GD on W with fresh samples across steps, then re-estimate the second layer on new data. This two-stage protocol is useful analytically, but differs from common pipelines that reuse minibatches, update (W, a) jointly, and use momentum or adaptive methods. The analysis also uses an auxiliary symmetrized initialization that sets the initial predictor to zero and simplifies the first update. As a result, the reported rates should be interpreted for this training protocol, not for arbitrary optimization settings.

What changes in the early-time feature-learning picture when initialization is not symmetrized?

What changes if (W, a) are trained simultaneously with separated time scales (larger step size for the first layer, smaller step size for the second layer), a setup closer to practical training than strict two-stage optimization, and can the paper’s conditioning/concentration framework be adapted to analyze this smoothed two-step regime?

Do adaptive optimizers such as Adam induce comparable layerwise time-scale separation in practice (or the opposite, if backpropagated gradients attenuate in earlier layers), and can training be accelerated by schedule designs inspired by this paper that avoid treating all layer parameters identically?

Can the early-time feature-learning phenomena analyzed here motivate alternatives to standard backpropagation (for example, forward-only local objectives such as the forward-forward algorithm (Hinton, 2022)), especially if one seeks biologically plausible update mechanisms?

Generalized-depth alignment question (exact setup): let the teacher be two-layer,

$$y = h^*(U^*g^*(V^*z)), \quad (3)$$

with $V^* \in \mathbb{R}^{r_1 \times d}$ (inner teacher directions), $U^* \in \mathbb{R}^{r_2 \times r_1}$ (outer teacher directions), inner nonlinearity $g^* : \mathbb{R}^{r_1} \rightarrow \mathbb{R}^{r_1}$, and outer map $h^* : \mathbb{R}^{r_2} \rightarrow \mathbb{R}$. Let the student be three-layer,

$$\hat{y} = a^T \sigma_2(W_2 \sigma_1(W_1 z)), \quad (4)$$

with $W_1 \in \mathbb{R}^{p_1 \times d}$ and $W_2 \in \mathbb{R}^{p_2 \times p_1}$. Under what conditions do we get staged alignment $\text{Row}(W_1^t) \rightarrow \text{Row}(V^*)$ and $\text{Row}(W_2^t) \rightarrow \text{Row}(U^*)$ during early training? Which geometric conditions (principal-angle / range-nullspace separation between teacher and student subspaces) and which complexity descriptors of g^*, h^* (e.g., separate leap indices, Hermite tensor ranks, derivative bounds) are sufficient for this behavior?

2.3 PRELIMINARY EXPERIMENTS ADDRESSING OPEN QUESTIONS

This subsection is an empirical check of two questions posed above. It does **not** add new proofs beyond (Dandi et al., 2024).

2.3.1 MOTIVATION AND HYPOTHESES

Scope. The simulations below do not prove new results. The single-step directional bottleneck and the role of batch scaling and leap structure are proved in (Dandi et al., 2024); we use simulations only to probe robustness under protocol variants (noise and initialization) that the paper does not treat in full generality.

2.3.2 WHAT IS SIMULATED (STEP-BY-STEP)

Student model. A two-layer network as in (Dandi et al., 2024):

$$\hat{f}(z; W, a) = \frac{1}{\sqrt{p}} \sum_{i=1}^p a_i \sigma(\langle w_i, z \rangle), \quad (5)$$

with ReLU σ , width p , and rows w_i of $W \in \mathbb{R}^{p \times d}$.

Data. Draw covariates $z \propto \mathcal{N}(0, I_d)$. Teacher directions w_j^* are sampled once per random seed; targets $f^{*(z)}$ are explicit polynomials in inner products $\langle w_j^*, z \rangle$ (single-index cases A/B in the first figures; two-index weighted quadratic in the third figure).

Training labels. Either clean $y = f^{*(z)}$ or noisy $y = f^{*(z)} + \varepsilon$ with $\varepsilon \propto \mathcal{N}(0, \sigma^2)$; we sweep $\sigma \in \{0, 0.05, 0.1, 0.2, 0.4\}$.

Initialization. Two protocols from (Dandi et al., 2024): Eq. (10) i.i.d. draws for (W_0, a_0) , versus Eq. (10)+(11) symmetrized pairing (duplicate hidden directions with opposite output weights).

Stage 1—first layer only (two ablations).

- **Frozen- W (random features baseline):** keep $W = W_0$; do **not** update the first layer.
- **One-step- W :** take **one** full-batch gradient step on W only, using squared loss on the **training** batch and **fixed** initial second layer a_0 (Alg. 1 style). Learning rate follows the paper-style scaling $\eta \propto p\sqrt{\frac{n}{d}}$.

Stage 2—readout. Refit a by ridge regression on the **same** training batch with W fixed at its Stage 1 value. This matches train a after an early-time move of W , “not joint training of (W, a) ”.

Evaluation. Report performance on a fresh Gaussian test set (same f^* , clean labels for test error). **Alignment metric:** $\max_i |\cos \angle(w_i, w_j^*)|$ over hidden rows—a concrete cosine overlap with teacher directions, comparable in spirit to measuring whether $\text{Row}(W)$ aligns with teacher subspaces as in (Dandi et al., 2024) §2.1 / Fig. 1, but not identical to any single quantity in their proofs.

Figure layout (single-index plots). Case A vs Case B form the two rows; columns cycle initialization $\times \sigma$. Within each subplot, two bars compare frozen- W vs one-step- W . The **Results** subsection states what each figure plots. Hyperparameters (d, n, p, η, λ) are held fixed across σ (no retuning per noise level).

Typical numerics (unless overridden when generating figures). Reference scripts use $d = 128$, training batch $n = 8d$, width $p = 512$, test sample size 4096, learning rate $\eta = 5 \cdot p\sqrt{\frac{n}{d}}$, and ridge readout λ on the order of 10^{-3} to 10^{-6} depending on the run.

Motivation of these experiments.

- **Noise-motivation test:** the question in §2 asks whether moderate label noise changes alignment between student rows and teacher directions beyond what follows from higher MSE alone, inspired by noise-as-regularizer and stochastic-resonance discussions (Bishop, 1995; Gammaioni et al., 1998).
- **Initialization-motivation test:** (Dandi et al., 2024) rely on symmetrized initialization for analysis. We test whether the observed early-time behavior is sensitive to Eq. (11), or whether the qualitative alignment pattern survives without that symmetry constraint.

2.3.3 RESULTS AND INTERPRETATION

The figures below summarize runs from the Stage 1–2 pipeline in the preceding subsection **What is simulated (step-by-step):** sample (W_0, a_0) , optionally take one gradient step on W with fixed a_0 under squared loss on noisy training labels, ridge-fit a , then evaluate on i.i.d. test draws from $\mathcal{N}(0, I_d)$ with clean targets $f^{*(z)}$ for MSE. Commands and paths to regenerate the plots are listed in the Appendix. Table 1 maps each panel to objects and results in (Dandi et al., 2024) (Eq., Alg., Thm.) versus what we measure on finite draws; the bullets below spell out subplot grids.

Table 1: Companion to the plots below. Rows 1–2 are one **experiment** with two summaries per replicate; row 3 uses a different teacher. A **seed** fixes the pseudorandom generator state for one replicate; changing it redraws (W_0, a_0) , teacher directions, training batches, and noise—each seed is one independent Monte Carlo trial. Bars average over several seeds; error bars indicate spread across those trials. Protocol objects (Eq. (1)–(2), Alg. 1 phases, Eq. (10)–(11), $\eta \propto p\sqrt{\frac{n}{d}}$) follow (Dandi et al., 2024); Hermite-style Case A/B targets and label-noise sweeps are exploratory. Alignment“ means $\max_i |\cos \angle(w_i, w_j^*)|$. No formal testing; (d, n, p, η, λ) fixed across σ .

Panel	Question (narrow)	Where the protocol is anchored in (Dandi et al., 2024)	Outcomes & caveats (finite Monte Carlo trials)
Single-index test MSE Figure 1	Does one GD step on W lower clean test MSE vs frozen- W , across Case A/B targets, init (Eq. (10) vs Eq. (10)+(11)), and training noise σ ?	Same model class and idealized training as (Dandi et al., 2024): predictor as in their Eq. (1)–(2); two phases as in Alg. 1 (here one optional gradient step on W with $a = a_0$, then refit a); learning rate chosen in the same scaling family as their Fig. 3 experiments ($\eta \propto p\sqrt{\frac{\pi}{d}}$). Initializations follow Eq. (10) and, when used, Eq. (10)+(11). Extension: Gaussian inputs and teacher structure match the paper’s setting; additive label noise on training data is an extrapolation —their theorems assume noiseless $y = g^{*(W^*z)}$ (Dandi et al., 2024).	Case A: one-step mean below frozen at every σ in the plot; smallest Case A error at $\sigma = 0$. Case B: smaller margins and stronger init dependence; one-step can exceed frozen under Eq. (11). No turn-key evidence that raising σ reduces test MSE (exploratory sweep, not tuned λ per σ).
Single-index alignment	Same Monte Carlo trials as Figure 1. Does the same step increase $\max_i \cos \angle(w_i, w^*) $ vs frozen- W ? (A second summary statistic, not a separate experiment.)	Identical trials as the preceding row. The cosine measures overlap between hidden rows and w^* , i.e. row-level overlap with the teacher direction—the same geometric object Thms. 4–5 / Fig. 1 discuss via alignment with (subspaces spanned by) teacher directions at finite batch step (Dandi et al., 2024); we report a concrete scalar diagnostic, not their asymptotic theorem conclusion.	In all panels of the saved figure, one-step alignment mean $>$ frozen mean at each tested σ . Training noise hurts test MSE (first panel) but does not reverse this cosine ordering in these panels.
Two-index alignment	With $r = 2$, does one step yield unequal alignment to w_1^* vs w_2^* compared with frozen- W ? This is motivated by Thms. 4–5 (Dandi et al., 2024): at batch scale $n = \Theta(d)$, a single giant step cannot fully specialize across multiple generic directions (their Fig. 1 illustrates the batch-threshold picture).	Alg. 1-style training and Eq. (10)–(11) inits as above (Dandi et al., 2024); explicit polynomial f^* with two orthonormal directions (an illustrative multi-index teacher, not one of the paper’s formal instantiations). Same (σ, init) comparison logic as row 1; figure layout differs from Figure 1.	In these plots $ m_1 - m_2 $ often increases after one step vs frozen under clean and noisy training labels; seed dependence remains. Interpreting strictly: Thms. 4–5 concern large- d limits and noiseless labels (Dandi et al., 2024), so this panel is only a finite-width illustration of asymmetric ⁴ alignment across directions after one step, not a numerical verification of their statements.

What each figure displays.

- **Single-index, test MSE.** Teacher uses one unknown direction w^* . Each subplot shows mean test MSE (error bars: variability across random seeds). **Rows:** Case A (quadratic / Hermite- H_2 -style target) and Case B (quartic / H_4 -style target). **Columns:** initialization—Eq. (10) i.i.d., and Eq. (10)+(11) symmetric when included—crossed with training-noise level $\sigma \in \{0, 0.05, 0.1, 0.2, 0.4\}$. **Bars:** frozen- W (left) vs one-step- W (right). $\Delta \text{MSE}'$ printed on each panel is percent change from the frozen mean to the one-step mean.
- **Single-index, alignment.** Same experimental grid as the test-MSE figure for the same seeds and hyperparameters; only the plotted statistic changes to $\max_i |\cos \angle(w_i, w^*)|$ from W after Stage 2 (the same W used for prediction).
- **Two-index teacher, alignment.** Same Stage 1–2 machinery, but f^* depends on two orthonormal directions (w_1^*, w_2^*) (formula in the figure caption). **Rows:** frozen- W vs one-step- W . **Columns:** initialization $\times \sigma$. Each subplot has two bars (alignment to w_1^* and to w_2^*). The value $|a_1 - a_2|$ printed on the plot is the absolute difference between those two reported means.

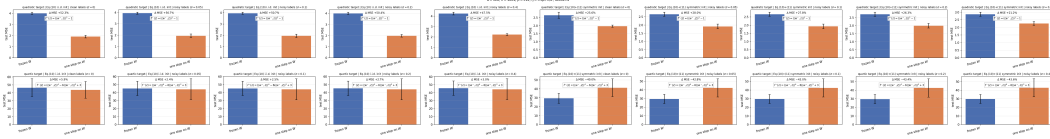


Figure 1: Single-index teacher: mean test MSE ($\pm$ variability across seeds). Rows: Case A / Case B. Columns: Eq. (10) vs Eq. (10)+(11) and training-label noise σ . Bars: frozen- W , one-step- W . $\Delta \text{MSE}'$ is percent change from frozen mean to one-step mean. **Figure interpretation (MSE).**

- Case A (quadratic): one-step improves over frozen- W for all tested σ , but best absolute error is at $\sigma = 0$.
- Case B (quartic): gains are weaker and more initialization-sensitive; under Eq. (11), one-step can worsen MSE.
- Overall: no evidence here that moderate noise improves test error.

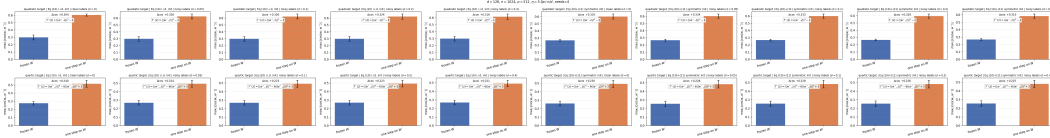


Figure 2: Same seeds and layout as Figure 1. Vertical axis: $\max_i |\cos \angle(w_i, w^*)|$ after ridge readout (same W as in Figure 1).

Figure interpretation (alignment). Each bar plots $\max_i |\cos \angle(w_i, w^*)|$ (maximum cosine similarity between a hidden row and the teacher direction w^*), i.e. the usual best neuron alignment score; this is not test MSE. For each subplot we compare two procedures at the same (σ, init) : **frozen- W** leaves the first layer at its initialization; **one-step- W** takes one full-batch gradient step on W before fitting a . In every panel the one-step bar is strictly larger than the frozen bar: the alignment score after one step exceeds the score with no step, at each tested noise level. So label noise worsens test error in the previous figure, but it does not reverse this ordering on the cosine statistic shown here.

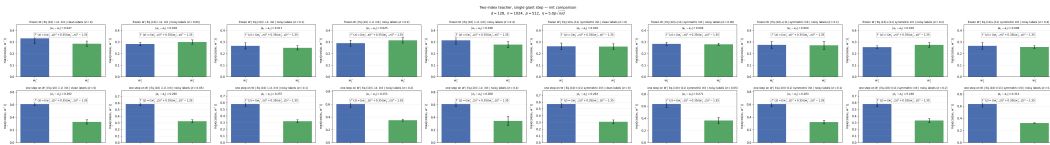


Figure 3: Two-index teacher, $f^{*(z)} = (\langle w_1^*, z \rangle)^2 + 0.35(\langle w_2^*, z \rangle)^2 - 1.35$ with orthonormal (w_1^*, w_2^*) . Rows: frozen- W vs one-step- W . Columns: initialization $\times$ training noise σ . Two bars per panel: alignment to each teacher direction; $|a_1 - a_2|$ is the absolute gap between those two means.

Figure interpretation (two-index teacher).

- Each panel has two bars: alignment to w_1^* and alignment to w_2^* , using the same cosine statistic as above. The printed $|a_1 - a_2|$ is the absolute difference between those two bar heights (not a second-layer weight difference).
- In the plotted means, the absolute difference between the two bar heights is often larger after one step on W than for frozen- W ; the same qualitative pattern appears with clean and with noisy training labels in these runs.
- Qualitatively, this matches the batch-size story in (Dandi et al., 2024) (Thms. 4–5, Fig. 1): at $n = \Theta(d)$ one giant step can align strongly with one teacher direction while leaving another weaker, rather than specializing evenly across both.

2.3.4 QUANTITATIVE SUMMARY AND LIMITS

Preliminary takeaway (not a theorem). In these finite-width runs, I do not observe a robust moderate-noise improves test error’’ effect. At each tested σ , one-step- W still yields higher cosine alignment than frozen- W in the plots above, while test MSE generally rises as σ increases. Symmetrized initialization changes the baseline predictor and can change measured one-step improvements; (Dandi et al., 2024) likewise stress that protocol choices affect such comparisons.

Quantified examples (mean $\pm$ std over 4 seeds; conditional on fixed hyperparameters). In the one-step single-index experiment with Eq. (10) i.i.d. init and $\sigma = 0$, Case A test MSE changes $4.01 + -0.11 \rightarrow 1.91 + -0.12$ (frozen to one-step). At $\sigma = 0.4$, Case A changes $4.25 + -0.14 \rightarrow 2.23 + -0.10$; this is not evidence of noise-induced improvement over the clean-label baseline. In the two-index setting, alignment to w_1^* and to w_2^* after one step remain unequal under both clean and noisy labels in these runs (quantified on the figure by $|a_1 - a_2|$), while reported MSE depends on σ .

Inference caveat. These are finite-width, finite-seed observations under fixed schedules; we do not re-optimize hyperparameters per noise level, and we do not perform formal hypothesis testing. Therefore, the noise-related statements should be interpreted as setup-conditional empirical observations.

2.4 TARGET GEOMETRY: REGULARITY, STAIRCASE, LEAP, OBSTRUCTED TARGETS

The main bounds use staircase/polynomial structure in g^* and leap index ℓ along each direction (Dandi et al., 2024). Figure 2 includes cases where no teacher direction is recovered in finite depth, and some directions require batch scaling $\Theta(d^\ell)$. These are algebraic constraints in Hermite space (which orders carry mass along which directions). This perspective is different from landscape-based analyses of non-convex optimization and does not directly cover deeper compositional architectures.

Do recently proposed non-backprop / random-search post-training approaches (e.g., random perturbation experts in (Gan and Isola, 2026) and hyperscale ES variants in (Sarkar et al., 2026)) admit a connection to staircase-style sequential feature recruitment, or do they rely on a different geometric mechanism?

Related jumping staircase’ question: if recovery of a new direction at Hermite order $k + 1$ depends on how many tensor factors are already shared with learned order- k structure, does the theory imply an intermediate regime where $n = \Theta(d^2)$ (rather than $\Theta(d)$) can unlock additional directions in one giant step when overlap is partial? If such jumps exist, can they motivate layerwise learning-rate or batch-size schedules that differ across layers instead of uniform adaptive updates?

2.5 CONCENTRATION REGIME: n OF ORDER d AND THE GAUSSIAN TOOLBOX

Operator-norm concentration for empirical covariances already requires n to scale with d when directions are unconstrained. This underlies the paper’s batch-size thresholds before leap/staircase refinements. The proof techniques (projection conditioning, Gaussian equivalence, Hermite calculus) are specialized to Gaussian covariates, so extension to non-Gaussian designs requires new arguments.

2.6 WHAT IS PROVED VS CONJECTURED (TABLE 1) + SCOPE IN DL THEORY

Table 1 in (Dandi et al., 2024) includes both proved results and entries labeled as educated guesses, so claims should be distinguished accordingly. Relative to broader theory, the paper focuses on finite-time, finite-width directional alignment in V^* , while NTK/lazy and mean-field lines emphasize different asymptotic objects. The structural contribution is the interaction of batch size, iteration depth, and Hermite geometry of g^* ; conclusions that depend on Gaussian noiseless inputs, fresh batches, and decoupled training should be read as model-specific.

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A APPENDIX

A.1 REPRODUCIBILITY NOTES FOR PRELIMINARY EXPERIMENTS

Scripts:

- `experiments/one_step_vs_rf.py`
- `experiments/two_index_one_step.py`

Representative commands used for the figures included in this report:

- `python experiments/one_step_vs_rf.py --paper --ridge 1e-6 --seeds 4 --noise-grid 0,0.05,0.1,0.2,0.4`
- `python experiments/two_index_one_step.py --seeds 4 --noise-grid 0,0.05,0.1,0.2,0.4`

Outputs:

- `experiments/figures/one_step_vs_rf_mse.png`
- `experiments/figures/one_step_vs_rf_alignment.png`
- `experiments/figures/two_index_one_step_alignments.png`